Anant Khanna

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Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science in Computer Engineering - Dean's List 2025 *Expected Graduation: 2028*

Experience

Founding Software Engineer | *Agriprix* May 2026 – Present
- Rebuilt the hybrid intent router (deterministic rules + LLM classification) for an agent platform spanning **70+** tools – reaching **96%** routing accuracy, **45%** lower median latency, and **25%** lower per-request token cost – with graceful fallback to deterministic rules on classifier timeout and per-call token/latency observability
- Implemented Stripe authorize-and-capture checkout in a Django/DRF backend deployed via Docker, with deterministic idempotency keys, an integer-cent commission pricing engine, and per-farm fulfillment state machines for multi-supplier orders
- Led a platform security audit surfacing **67** vulnerabilities (5 critical): patched mass-assignment/IDOR flaws in Django REST Framework serializers, eliminated overselling races via row-level locking, and shipped a 15-step end-to-end Playwright journey covering the supplier-to-buyer order flow

Machine Learning Software Engineer Team Lead | *UBC UAS* Sept. 2024 – Present
- Integrated YOLOv11 (PyTorch) pipeline in Python and C++, enabling real-time autonomous object detection on embedded systems travelling at **50+** km/h, optimizing inference for low-latency execution
- Implemented TensorRT FP16 quantization to compress model weights, reducing memory bandwidth usage by **50%** while maintaining inference precision within **0.5%** of the baseline

Co-founder & Lead Developer | *UBC Finds* Oct. 2025 – May 2026
- Built a campus utility finder map on React/Next.js and Supabase, architecting event-driven state synchronization over Supabase realtime channels that eliminates race conditions through atomic writes

Projects

Attention-Aware Robot Lamp | *Python, MediaPipe, YOLO11, Whisper, LLM tool calling* Aug. 2026
- Built a real-time gaze-engagement detector for a 6-DOF robot lamp using MediaPipe face landmarks with two-threshold + dwell-frame hysteresis, reaching 0.952 windowed F1 (0.933 precision / 0.972 recall) on a **1,224**-frame labeled session at 13.8ms median frame latency
- Designed an LLM tool-calling agent (Anthropic/OpenAI) over a persistent object memory fed by interval-gated YOLO11 scans, exposing recall, reminders, light control, and live vision judgments as tools – full voice round trip (Whisper wake-word transcription in ~155ms chunks to spoken reply) in 4-6 seconds

GPU Inference Benchmark | *Python, PyTorch, CUDA/TensorRT, ROCm/MIGraphX* July – Aug. 2026
- Built a three-rung benchmark ladder (PyTorch -> TensorRT FP32 -> FP16) that holds precision constant across stages to isolate graph-compilation gains from quantization gains, benchmarking YOLO11n on an RTX **3080** across **100** timed runs per config
- Ported the harness to AMD MI300X via ROCm/MIGraphX, finding unoptimized PyTorch FP32 on the MI300X (**1,143** fps) already beats the fully-optimized TensorRT FP16 engine on the RTX **3080** (**1,008** fps) through raw hardware scale
- Scaled the MIGraphX FP16 engine from batch 1 to 8 with near-zero latency penalty (1.03ms -> 1.10ms), reaching **7,277** fps by saturating the MI300X's memory bandwidth

Wealth Council | *Python, FastAPI, React, Google Gemini, Fetch.ai* Mar. 2026
- Engineered an agentic investment platform, winning an award at the ProduHacks x Fetch.ai hackathon
- Architected a Multi-Agent Orchestration Layer using Fetch.ai uAgents to asynchronously compute portfolio risk metrics and integrated FinBERT (NLP) to perform sentiment analysis on live financial news streams

Technical Skills

Languages: Python, C, C++, C#, SQL, Java, JavaScript, TypeScript, RISC-V & ARM Assembly
AI/ML: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Generative AI, Conversational AI, Transformers, AI Evaluation, Intent Recognition
Frameworks: Django/DRF, FastAPI, Flask, Celery, React, Next.js, PyTorch, TensorFlow, YOLO/Ultralytics, NumPy, Matplotlib, REST APIs
Tools & Cloud: Google Cloud Platform, Docker, PostgreSQL, MySQL, Supabase, Stripe, Playwright, Git, Linux
Hardware/RTL: FPGA Design (Intel Quartus, QuestaSim, Platform Designer), Verilog, SystemVerilog, ARM SoCs